

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1706

Weyerhaeuser Pine Hill Operations 138 kV
May 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1706. The Interconnection Customer proposes a 331.3 MW large load (data center) facility interconnecting at the Weyerhaeuser Pine Hill Operations 138 kV substation in the Maryland area. The request was received on March 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 568 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,997,000.

3. Network Impacts

3.1 Load Deliverability Analysis

For the loss of RIVA ROAD 138-1 kV XFMR, the SILOAM SPRINGS SOUTHWEST / EDEN PRAIRIE 138.0 kV loads to 173.1% of its 258 MVA rating. Project DFAX: 100.0%.

For the loss of RIVA ROAD 138 kV/1 kV Transformer, the LAYLAND / ADRIAN ENERGY ASSOCIATES LLC 138.0 kV line loads to 166.2% of its 245 MVA rating. Project DFAX: 5.7%.

The SILOAM SPRINGS SOUTHWEST / LITTLETON 138.0 kV Line is overloaded to 163.9% of its 261 MVA emergency rating for the loss of the RIVA ROAD 138 kV/1 kV Auto. The project contributes 100.0% DFAX (331.3 MW) to this overload.

JEROME / LAYLAND 138.0 kV: 157.9% loading (235 MVA) following the outage of the RIVA RD 138/1 kV Transformer Bank; DFAX = 5.7%.

For the loss of RIVA RD 138-1 kV Transformer, the LITTLETON / WEYERHAEUSER PINE HILL OPERATIONS 138.0 kV loads to 154.4% of its 264 MVA rating. Project DFAX: 100.0%.

For the loss of RIVA RD 138/1kV TX, the BELTSVILLE / PRAIRIE WIND FARM 138.0 kV Line loads to 154.2% of its 191 MVA rating. Project DFAX: 43.1%.

The WEYERHAEUSER PINE HILL OPERATIONS / RIVA RD 138.0 kV Circuit 1 is overloaded to 138.5% of its 266 MVA emergency rating for the loss of the RIVA RD / WEYERHAEUSER PINE HILL OPERATIONS 138.0 kV Cir. 2. The project contributes 65.6% DFAX (217.4 MW) to this overload.

For the loss of RIVA RD / WEYERHAEUSER PINE HILL OPERATIONS 138.0 kV Ckt 1, the RIVA RD / WEYERHAEUSER PINE HILL OPERATIONS 138.0 kV #2 loads to 138.5% of its 266 MVA rating. Project DFAX: 65.6%.

CHALK POINT / HOLLAND CLIFF 138.0 kV: 108.2% loading (237 MVA) following the outage of the RIVA ROAD SUBSTATION / SHINGLETOWN 345.0 kV; DFAX = 5.5%.

The WHITE PLAINS / HOLLAND CLIFF 138.0 kV line is overloaded to 103.8% of its 233 MVA emergency rating for the loss of the RIVARD / SHINGLETOWN SUBSTATION 345.0 kV Line. The project contributes 5.5% DFAX (18.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EDEN PRAIRIE / SILOAM SPRINGS SOUTHWEST 138.0 kV	Rebuild line with higher-capacity conductor	\$6,942,000	100% (\$6,942,000)
LAYLAND / ADRIAN ENERGY ASSOCIATES LLC 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,312,000	92% (\$1,211,000)
LITTLETON / SILOAM SPRINGS SOUTHWEST 138.0 kV	Rebuild line with higher-capacity conductor	\$1,433,000	100% (\$1,433,000)
JEROME / LAYLAND 138.0 kV	Rebuild line with higher-capacity conductor	\$10,129,000	92% (\$9,346,000)
LITTLETON / WEYERHAEUSER PINE HILL OPERATIONS 138.0 kV	Rebuild line with higher-capacity conductor	\$2,052,000	100% (\$2,052,000)
BELTSVILLE / PRAIRIE WIND FARM 138.0 kV Line	Rebuild line with higher-capacity conductor	\$9,700,000	88% (\$8,573,000)
WEYERHAEUSER PINE HILL OPERATIONS / RIVA RD 138.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,579,000	100% (\$1,579,000)
RIVA RD / WEYERHAEUSER PINE HILL OPERATIONS 138.0 kV	Rebuild line with higher-capacity conductor	\$1,271,000	100% (\$1,271,000)
HOLLAND CLIFF / CHALK PT 138.0 kV	Reconductor line	\$1,648,000	100% (\$1,648,000)
HOLLAND LIFF / WHITE PLAINS 138.0 kV Line	Reconductor line	\$942,000	100% (\$942,000)
	Total allocated to project		\$34,997,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.